

AMANDI NIMASHA

SOFTWARE ENGINEER

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Dedicated Software Engineer with experience in **Full Stack Development, Machine Learning, and Web3 Technologies**. Skilled in building scalable applications with experience leading development teams and delivering production-ready solutions. Strong research background with the ability to quickly adapt to new technologies and solve complex problems.

SKILLS

Languages

TypeScript, JavaScript, Python, Java, SQL, .NET (C#), HTML5, CSS3

Frontend Development

React, Next.js, Angular, React Native, Vite TailwindCSS, Angular Material, Streamlit

Backend & Frameworks

NestJS, ExpressJs, FastAPI, Node.js, .NET 10

Databases & Storage

PostgreSQL, MongoDB, SQL Server, SQLite, Redis, Firebase, Convex

Cloud & Infrastructure

Google Cloud Platform (GCP), Docker, Nginx, Prometheus, Grafana,

Web3 Technologies

Bitcoin Computer, Bitcoin Ordinals/Inscriptions, Solana, Ethereum

Tools & Methodologies

Git/GitHub, CI/CD (GitHub Actions, GitLab CI/CD), n8n (Automation), Agile/Scrum, Unit & Integration Testing (Jest, Mocha/Chai), Automated Testing,

CERTIFICATES

Docker - swarm-service-stack - Hands-on, KodeKloud (Issued Feb 2026)

Amazon Elastic Container Service (AWS ECS)

Agile Development and Scrum, IBM (Issued Dec 2024)

GitHub Actions Certification, KodeKloud (Issued Sep 2025)

GitLab CI/CD: Architecting, Deploying, and Optimizing Pipelines, KodeKloud (Issued Sep 2025)

Multi AI Agent Systems with crewAI, DeepLearning.AI (Issued Mar 2025)

WORK EXPERIENCE

Metarune Labs

Software Engineer

JUN 2025- PRESENT

- **Led a cross-functional engineering team** to successfully deliver multiple high-impact Web3, live-streaming, and fintech applications from conception to production-ready deployment.
- **Engineered decentralized Web3 systems and smart contracts** utilizing Bitcoin Computer and the TBC framework for secure on-chain token economies, asset minting, and royalty enforcement.
- **Integrated complex monetization and payment infrastructures**, including Stripe, PayPal, and real-time Lightning Network microtransactions.
- **Automated company internal workflows** and repetitive manual processes by architecting custom third-party integrations using n8n.
- **Designed and implemented scalable backend architectures** and RESTful APIs using NestJS to support real-time chat, creator dashboards, and multi-step workflows.
- **Built highly responsive, production-ready user interfaces** and cross-platform mobile apps using React, Next.js, and React Native, tightly integrating complex third-party media and mapping APIs.

Associate Software Engineer

JUN 2024- JUL 2025

- **Collaborated in Agile development cycles** utilizing Jira for sprint planning, task tracking, and continuous delivery of feature pipelines.
- **Built high-performance, responsive user interfaces** and intuitive dashboards with Next.js and TypeScript, optimizing the overall platform UI/UX.
- **Designed and implemented scalable backend APIs** using NestJS to support token-gated subscriptions and NFT-based ticketing architectures.

Software Engineer Intern

DEC 2023- MAY 2024

- **Developed and optimized frontend UI components** using Next.js, TypeScript, and TailwindCSS for Chainwork, a decentralized service marketplace.
- **Built and implemented core platform features** designed to streamline user workflows, matchmaking, and interactions on the platform.
- **Engineered real-time chat functionality** by integrating WebSockets to facilitate seamless, live communication between clients and service providers.

EDUCATION

BSc. in Engineering (Hons), Computer Science and Engineering

2021 - 2025

University of Moratuwa, Sri Lanka

Second Class Upper Division , **Addition:** Minor degree in Mathematics

VOLUNTEERING

Editor, IEEE Signal Processing Society Student Branch, University of Moratuwa (Oct 2023 - Jan 2025)

Assistant Event Organizer, IEEE Signal Processing Society Student Branch, University of Moratuwa (Jan 2023 - Oct 2023)

PROJECTS

Heystream - Live Streaming Platform

- Built a full-stack live streaming platform from scratch using Next.js and NestJS in a Turborepo monorepo.
- Implemented real-time features with Ably, live video delivery via Livepeer, authentication with Clerk, subscription/payment flows with Stripe, and creator-to-viewer chat with TalkJS.
- Tech stack: **Next.js, React, TypeScript, NestJS, Prisma, Tailwind CSS, Livepeer, Stripe, Ably, Clerk, PostgreSQL**

React Gaming Platform - Playverse

- Built a full-stack gaming platform with a catalog of playable browser games including an endless runner, maze navigator, card game, and action shooter.
- Integrated Clerk authentication with a custom gaming-themed UI, supporting sign-in/sign-up modals and protected routes.
- Implemented a centralized game registry with filtering, sorting and smooth GSAP scroll animations.
- Tech stack: **React, TypeScript, Vite, Tailwind CSS, GSAP, EmailJS**

Real-Time Chat & Video Calling App

- Built a full-stack real-time communication platform utilizing Angular 19 and .NET 10 to support instant messaging alongside peer-to-peer video and voice calls.
- Implemented P2P video and voice communication using WebRTC, configuring STUN server negotiation and ICE candidate queuing via SignalR signaling hubs.
- Developed JWT-authenticated SignalR hubs to manage real-time messaging, typing indicators, read receipts, active online presence states, and cursor-based message pagination.
- Designed a database schema to log comprehensive call metadata, capturing session duration, multi-state status tracking (completed, missed, declined, cancelled), and call direction flags per user.
- Tech stack: **Angular, TypeScript, ASP.NET Core, .NET 10, WebRTC, SQLite, TailwindCSS, Docker**

AG Marketplace - Mobile automotive marketplace

- Implemented vehicle listing, search filters, real-time messaging, and notifications.
- Developed multi-step listing creation flow with Cloudinary media management.
- Integrated Google Maps for location-based search and discovery.
- Tech stack: **React Native, Expo, TypeScript, Convex, Clerk, Google Maps API**

Kiddo Cart - School Van Service

- Developed a unified React Native (Expo) mobile app serving both Driver and Parent roles from a single codebase, featuring live GPS location tracking, attendance event recording, absence management, and a notification history feed.
- Engineered a NestJS REST API with role-based JWT authentication, Supabase (PostgreSQL) for data persistence.
- Tech stack: **React Native, Expo, TypeScript, NestJS, Supabase, Zustand**

LLM-based Travel Planner

- Designed a hierarchical multi-agent system using GPT-4o for personalized travel itineraries.
- Implemented supervisor-agent orchestration coordinating calendar, data retrieval, planning, and feedback agents.
- Built generate-evaluate-refine pipeline with rule-based scoring for constraint-aware itinerary optimization.
- Integrated APIs and datasets for flights, hotels, restaurants, and attractions with LangChain tools.
- Tech stack: **Python, LangChain, Langgraph, GPT-4o, LLM, Multi-Agent Systems**

Tumor.ai - AI-Powered Medical Imaging Platform

- Built a clinical decision support web application that empowers healthcare professionals with AI-powered detection of brain tumors, breast cancer, and skin cancer from MRI and medical images.
- Developed a responsive React/TypeScript frontend with role-based access, admin verification workflows, scan history tracking, and AI-generated heatmap visualizations.
- Integrated a Flask backend serving deep learning models (UNet, ResNet50) with Firebase authentication and MongoDB for data persistence.
- Tech stack: **React, TypeScript, Vite, Tailwind CSS, Flask, Python, TensorFlow, PyTorch, MondoDB, Firebase, ONNX Runtime, OpenCV**